🔍 What is Nmap?

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- Nmap (Network Mapper) is a command-line tool for discovering hosts and services on a network.
- Used for port scanning, service/version detection, OS fingerprinting, and scripted checks (NSE).
- Common in security, auditing, and network troubleshooting.

🔎 Host discovery

- Process of finding which IPs/hosts are alive before scanning them.
- Methods: ICMP ping, TCP/ARP probes, or -Pn to skip discovery.
- **Important: if hosts don’t respond to ping, Nmap may treat them as down.**

🔌 Port

- A numeric endpoint on a host (0–65535) where a service listens (e.g., 80 → HTTP).
- Ports map network traffic to applications/services.
- Scanning ports reveals which services are reachable.

✅ Open / ❌ Closed / 🚫 Filtered

- Open: Service is listening and will accept connections.
- Closed: Port reachable but no service listening (useful info).
- Filtered: Probes blocked or dropped by firewall (Nmap can’t tell if port is open).

🔀 TCP vs UDP

- TCP: Connection-oriented (handshake), reliable used by SSH, HTTP.
- UDP: Connectionless, simpler and often used for DNS, SNMP; harder to scan and noisier to analyze.
- Nmap handles TCP and UDP differently because of protocol behavior.

🧭 Scan types

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- Connect scan (-sT)
- SYN scan (-sS)
- TCP ACK scan (-sA)
- UDP scan (-sU)
- Version detection (-sV)
- OS detection (-O)
- Timing templates (-T0 ... -T5)
- Output formats (-oN, -oX, -oG, -oA)